configuration of the GPIO pins for the LEDs are controlled by bits 20 (red), 21 (yellow), and 22 (green) in the 32-bit GPDR0 register. These registers are memory mapped, thus they're easily accessible in C. We can read and write to these registers as if we are reading or writing from a memory location. This is what the ledInit function would look like.

```
#define PIN22_FUNC_GENERAL          (0xFFFFCFFF)
#define LED_GREEN                   (0x00400000)

void ledInit(void) {
        GPIO_0_CLEAR_REG = LED_GREEN;
        GPIO_0_FUN_HI_REG &= PIN22_FUNC_GENERAL;
        GPIO_0_DIRECTION_REG |= LED_GREEN;
}
```

Next up is the ledToggle function which turns it on and off.

```
void ledToggle(void) {
        if (GPIO_0_LEVEL_REG & LED_GREEN)
                GPIO_0_CLEAR_REG = LED_GREEN;
        else
                GPIO_0_SET_REG = LED_GREEN;
}
```

And finally to make it blink the delay.

```
#define CYCLES_PER_MS               (9000)

void delay_ms(int milliseconds) {
        long volatile cycles = (milliseconds * CYCLE_PER_MS);
        while (cycles != 0)
                cycles--;
}
```

The four functions main, ledInit, ledToggle and delay_ms control the blinking LED program. We'll discuss the building and execution of the program in the next section.